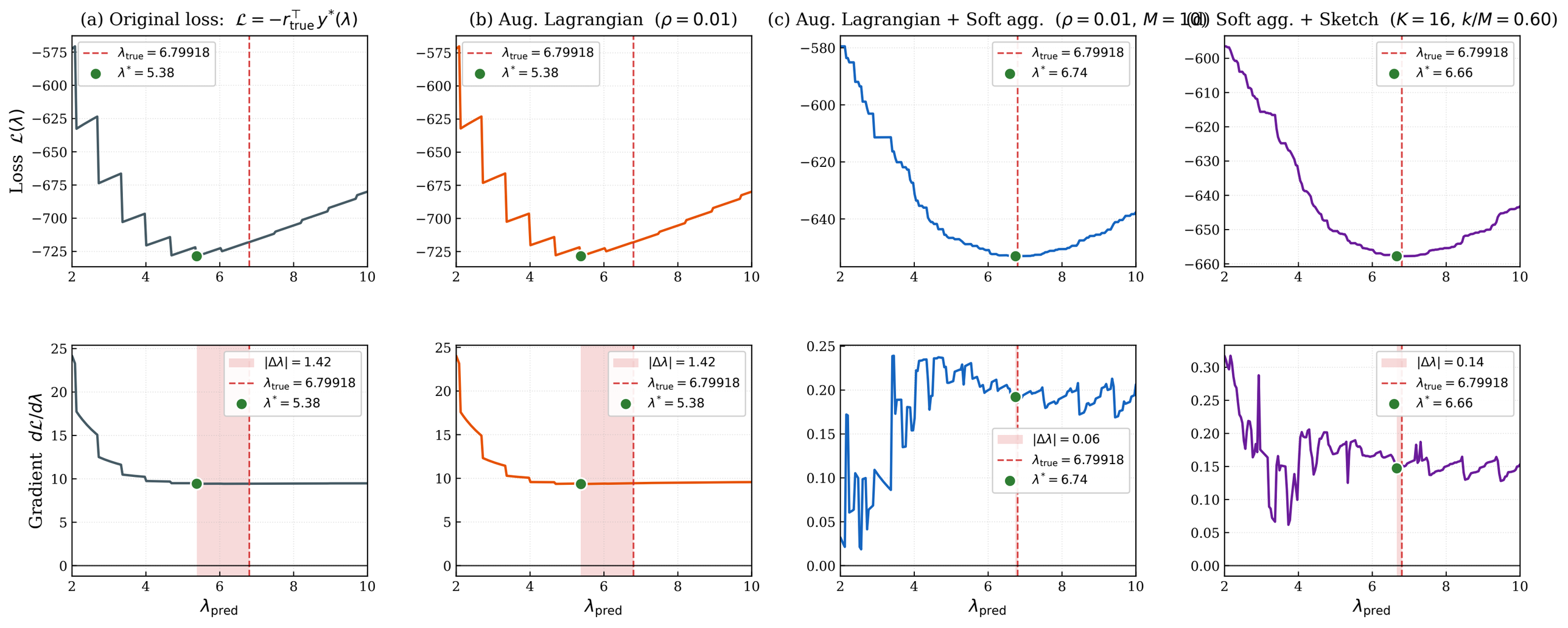
(a) Original loss: $\mathcal{L} = -r_{\text{true}}^\top y^*(\lambda)$
(b) Aug. Lagrangian ($\rho = 0.01$)
(c) Aug. Lagrangian + Soft agg. ($\rho = 0.01$, $M = 100$)
(d) Soft agg. + Sketch ($K = 16$, $k/M = 0.60$)
Loss $\mathcal{L}(\lambda)$
Gradient $d\mathcal{L}/d\lambda$
λ_{pred}
$\lambda_{\text{true}} = 6.79918$
$\lambda^* = 5.38$
$\lambda^* = 6.74$
$\lambda^* = 6.66$
$|\Delta\lambda| = 1.42$
$|\Delta\lambda| = 0.06$
$|\Delta\lambda| = 0.14$